Part C – Hands-on Exercises

13. Participant exercise distribution

Day 2 exercise bank

10

Easy

Collection fundamentals

10

Medium

Combined operations

5

Hard / Tricky

Complexity and edge cases

1

Integrated

Business application

26

Total exercises, with trainer solution material.

Easy – 10 exercises

No.	Exercise	Concepts
E01	Employee Name Registry	List creation and iteration
E02	Score Access and Slicing	Indexing, negative indexing and slicing
E03	Employee Attendance Counter	`count()`, `index()`
E04	Employee Name Sorting	`sort()` and `sorted()`
E05	Training Configuration	Tuple creation and unpacking
E06	Duplicate Employee ID Removal	Set conversion
E07	Common Technical Skills	Set intersection
E08	Employee Profile Dictionary	Dictionary creation and access

E09	Optional Employee Attributes	`get()` and default values
E10	Passing Score Collection	List comprehension and filtering

Medium – 10 exercises

No.	Exercise	Concepts
M01	Employee Record Management	List of dictionaries, append and update
M02	Employee Skill Frequency Counter	Dictionary counting pattern
M03	Duplicate Name Detection	Sets and ordered deduplication
M04	Shared Skills Analysis	Set union, intersection and difference
M05	Merge Training Configuration	Dictionary update and merge
M06	Department Employee Filtering	List comprehension
M07	Department-wise Performance Statistics	Dictionary and counters
M08	Employee ID and Score Mapping	`zip()` and dictionary comprehension
M09	Employee Report Generation	`enumerate()` and formatting
M10	Multi-criteria Employee Ranking	`sorted()` and tuple-based sorting

Hard / Tricky – 5 exercises

No.	Exercise	Challenge
H01	Deep Copy Employee Records	Shallow copy versus deep copy
H02	Safe Employee Record Removal	Avoid modifying collections during iteration
H03	Employee Lookup Error Handling	Missing keys and duplicate identifiers
H04	Deduplicate Records by Employee ID	Preserve the latest record without losing ordering
H05	Flatten and Analyze Nested Assessment Data	Nested comprehensions and aggregation

Integrated scenario – 1 exercise

Final practical challenge

H06 – Enterprise Employee Training Analytics System

A company has 100 employees across IT, HR, Finance and Operations departments. Each employee can have multiple skills and assessment records.

Develop an in-memory system that:

1. Stores employee information as a list of dictionaries.
2. Maintains a unique employee lookup using a dictionary.
3. Maintains unique departments and skills using sets.
4. Stores fixed certification criteria in a tuple.
5. Identifies eligible and ineligible employees.
6. Identifies duplicate IDs and duplicate names.
7. Finds common skills across departments.
8. Calculates department-wise average scores.
9. Generates top-five performers.
10. Creates a dictionary of employee IDs and scores.
11. Generates a report using `enumerate()`.

12. Uses comprehensions wherever they improve readability.

Constraints

- No OOPS or class definitions.
- No user-defined functions; these are introduced on Day 3.
- Validate the input records.
- Handle missing optional fields.
- Avoid modifying records during active iteration.
- Use only built-in Python collections and modules for the exercise.
- Run Ruff and Bandit before submission.

Expected outcome: A small but meaningful in-memory employee analytics application, ready to be refactored into a reusable application on Day 3.